

MUHAMMED RISWAN P

Data Scientist | ML Engineer

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Professional Summary

Data Science professional with 10+ months of hands-on experience building end-to-end machine learning pipelines and MLOps systems. Proficient in Python, Scikit-learn, PyTorch, and the full MLOps stack including Docker, FastAPI, GitHub Actions, DVC, and MLflow. Experienced in computer vision (YOLOv5, U-Net, face recognition), classical supervised and unsupervised ML, and production-grade REST API deployment. Seeking Junior Data Scientist/ML Engineer roles to deliver scalable, data-driven solutions.

Technical Skills

Languages: Python, SQL (PostgreSQL)

Machine Learning: Scikit-learn, Linear & Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, XGBoost, K-NN, Ridge & Lasso Regression, SMOTE, Cross-Validation, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV)

Unsupervised Learning: K-Means Clustering, Hierarchical Clustering, DBSCAN, PCA, Dimensionality Reduction, Elbow Method, Silhouette Score

Deep Learning: PyTorch, Artificial Neural Networks, CNNs, Transfer Learning (ResNet18, VGG16, MobileNet), Dropout, Batch Normalization, Adam & RMSProp Optimizers

Natural Language Processing (NLP): Hugging Face Transformers, BERT, DistilBERT, RoBERTa, Tokenization, Sentence Embeddings (SBERT), Semantic Similarity, Cosine Similarity, Transformer Architecture, Self-Attention, Encoder-Decoder Models, Hugging Face Pipelines, Text Summarization

Computer Vision: YOLOv5 (Object Detection), U-Net (Image Segmentation), OpenCV, Face Recognition, Grad-CAM, Inference Optimization

MLOps: Git, DVC, MLflow, Docker, FastAPI, Flask, GitHub Actions (CI/CD), EvidentlyAI, Cloud Deployment (Render)

Analytics & BI: Pandas, NumPy, Matplotlib, Seaborn, Power BI, DAX, Power Query

Statistics: Hypothesis Testing (Z-Test, T-Test, Chi-Square, ANOVA), Bayesian Statistics, Confidence Intervals, Correlation Analysis

Other: BeautifulSoup, Selenium, EDA, Feature Engineering, Data Wrangling

Experience

Data Science Intern

Jul 2025 – Present

Bridgeon Solutions, Kozhikode, Kerala

- Completed a structured 26-week intensive bootcamp covering statistics, SQL, Python, machine learning, deep learning, computer vision, and MLOps through hands-on projects.
- Designed and deployed an end-to-end MLOps pipeline using DVC, MLflow, Docker, FastAPI, and GitHub Actions with a live cloud REST API on Render serving real-time ML cluster predictions.
- Implemented automated CI/CD workflows and weekly data drift monitoring using EvidentlyAI with KS-test p-value alerting across 25 features.
- Trained and evaluated 15+ machine learning models across supervised, unsupervised, and deep learning paradigms with MLflow experiment tracking.

Projects

Customer Segmentation MLOps Pipeline

GitHub

Python • KMeans • PCA • FastAPI • Docker • DVC • MLflow • GitHub Actions • EvidentlyAI

- Engineered 3 customer segments from 2,240 records (29 features) using KMeans (k=3) and PCA retaining 90% variance.
- Built a complete MLOps stack with DVC, MLflow, FastAPI, Docker, and GitHub Actions.
- Deployed a live REST API on Render with automated CI/CD workflows and retraining pipelines.
- Implemented dual monitoring using EvidentlyAI reports and KS-test-based p-value alerting across 25 features.

Waste Classification CNN (WasteCNN)

PyTorch • CNN • MLflow • DVC • Docker • GitHub Actions

- Built a 6-class garbage image classifier achieving approximately 76% validation accuracy using a custom CNN architecture.
- Structured the project as a modular **src/** package following production-grade practices.
- Implemented MLflow experiment tracking, DVC versioning, Docker containerization, and GitHub Actions CI/CD.

Bank Marketing Subscription Prediction

GitHub

Python • Random Forest • Scikit-learn • Flask • Streamlit • Docker • GitHub Actions

- Predicted term deposit subscriptions on 41,188 records with severe class imbalance.
- Evaluated 8 machine learning models and selected a tuned Random Forest achieving ROC-AUC of 0.806.
- Built a complete Scikit-learn Pipeline with preprocessing and leakage prevention.
- Deployed a Flask API, Streamlit dashboard, and Dockerized application with CI/CD automation.

Olist E-Commerce Sales Performance Dashboard

GitHub

Power BI • DAX • Power Query • Business Intelligence

- Developed an interactive 4-page Power BI dashboard analyzing 100,000+ orders.
- Created KPI metrics and drill-down reports covering sales performance, delivery operations, customer satisfaction, and seller analytics.
- Identified delivery estimation issues through root-cause analysis and provided executive-level recommendations.

Education

B.Sc. Electronics

Graduated: 2025

WMO Arts and Science College, Muttill

University of Calicut | CGPA: 6.0